

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

TherAlign Health, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Founding engineer building full-stack features (Next.js + Firebase/GCP) end to end on a provider-facing AI product, iterating fast from prototype toward pilot in an early-stage environment.
- Extending a **Firebase Cloud Functions** backend that grounds an LLM (**Gemini**) in **RxNorm** normalization and PubMed/guideline evidence retrieval, with the LLM constrained to synthesis while the backend deterministically controls outputs, a grounded, evidence-first AI design.

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building an **NLP** / embeddings platform (Python, **FastAPI**, sentence-transformers, React), productionizing a published method into a self-serve tool and authoring its technical design doc and implementation guides.
- Built a YAML model registry, a versioned construct library (**99 constructs from 38 questionnaires**), and a structured data-quality warning system with stable machine-readable codes; grew the backend suite to **40 tests** (hermetic CI).

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant enterprise **SaaS** backend (Node.js, MongoDB) for **8+ UK clients**, processing **10,000+ monthly submissions** via REST API integrations; contributed to **\$340K+** in revenue.
- Architected a full-stack ML training-data annotation platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual annotation by **100+ hours/month** and lifting labeling throughput by **60%** as internal AI tooling.
- Piloted a COO-sponsored **GitHub Copilot** productivity experiment across code migrations, tech-debt PRs, and security-finding triage, maintaining a versioned conventions file to keep AI-generated code aligned with team patterns.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a FastAPI service that turns open-ended queries into structured entity tables via typed retrieval planning, reranking, and deterministic extraction with LLM fallback and **cell-level provenance**; hardened with **150+ tests** and provider fallback across OpenAI and Groq.

KG2RAG-Enhanced – Multi-Hop QA on HotpotQA (GitHub) Python, LlamaIndex, sentence-transformers, RRF

- Built the multi-view retrieval component of a **RAG** pipeline (query-side decomposition, embeddings, RRF fusion, cross-encoder reranking, MMR), lifting supporting-fact recall **+2.68%** over the KG²RAG baseline on a 4,905-question evaluation.

AutoEval – Retrieval Evaluation Improvement Agent (GitHub) Python, IR evaluation, Git automation

- Built an evaluation-improvement agent that proposes queries and labels, scores against multiple retrieval systems, and accepts only updates that raise discriminative power, with anti-Goodharting **guardrails** (schema checks, duplicate rejection, read-only holdout) and Git-backed report cards.

Publications

- “Alzheimer’s Disease Classification using Transfer Learning,” IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: Python, TypeScript, JavaScript, SQL, Java, C++, Bash, PostgreSQL, MongoDB, SQLite

AI & LLM: LLM agents, RAG, prompting, embeddings, sentence-transformers, Hugging Face Transformers, LlamaIndex, PyTorch, evaluation

Retrieval / RAG: dense retrieval, RRF, cross-encoder reranking, MMR, guardrails, cell-level provenance, Brave Search

Backend & Web: FastAPI, Node.js, Express, REST APIs, async services, React, Next.js, integration testing

Cloud & DevOps: Git, GitHub Actions, CI/CD, Docker, AWS, GCP, Azure, Linux

Open Source & Community

- Built **Underdogs Fitness** (Website), a production MERN platform with Stripe payments; live **3+ years**. - Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.